JPX Model Research Log

Three experiments | Reported results, hypotheses, and next steps

01 Ensemble: remove ReLU

Setup. Use the cited paper's lowest-average-MAE model. The baseline already removes the final prediction-layer ReLU to retain negative values for short-stock ranking. The next test removes all remaining ReLUs, including the one after the 128-unit linear layer.

Hypothesis. Removing ReLU may preserve information for later linear layers, especially negative predictions. GRU, LSTM, and RNN components still provide nonlinearity.

Metric	Baseline	All ReLUs removed
Unannualized Sharpe	0.0768	0.14462
Gross additive spread sum	17.78 pp	35.64 pp
Mean daily Rank IC	0.01899 (~0.019)	0.00508 (-73.2%)
Days with defined Rank IC	548	957 (+74.6%)

Interpretation. Sharpe nearly doubled and gross spread roughly doubled. More days had defined Rank IC as identical predictions became less common. The lower mean Rank IC may reflect newly included, harder-to-predict days rather than worse predictability.

Follow-up finding. The long leg drove the main gain and contributed 75.3% of gross additive spread sum, contradicting the hypothesis that gains mainly came from more detailed negative predictions and shorting. Luck or overfitting remains possible.

Validation and next steps

Execution-output checks. Compare original and no-ReLU long/short profits. Check Rank IC for the top 200 and bottom 200 stocks, and the percentage of actual top/bottom 200 stocks captured by predictions. The stated focus is on these extremes, rather than the middle ranks.

Proposed criteria. Gains not mainly from shorting would reject the negative-detail explanation. Rank IC for the top/bottom 200 below overall Rank IC was proposed as evidence of luck.

Out-of-sample plan. The long and short sides may have hard-to-observe microstructure. Obtain Japanese stock data for 2000-2016, clean it into JPX `stock_price.csv` format, and run each validation period on GPU using the model's final weights.

Proposed rejection rule. More than 10% of periods have a negative additive spread sum.

02 CNN-LSTM: remove ReLU and pooling

Test. Remove ReLU and pooling to test whether discarding values harms prediction.

Results. Daily Rank IC +27%; unannualized pooled Sharpe -45%; gross additive spread sum -9.6%.

Finding. The outcome was worse than the original CNN-LSTM. Removing model details did not necessarily improve performance.

03 Add CNN before Ensemble

Hypothesis. Add local features before the time-dependent model to gain more information from the data.

Results. Gross additive spread sum fell. Early stopping ended 15 of 17 training periods at epoch 6.

Interpretation. Low patience may have limited training, so the result does not establish that CNN-Ensemble is worse. The author considered `min_change` small (about 5% of average loss), and unlikely to be the main cause. Alternatively, the CNN layer itself may explain the decline.

Validation / next step. Set patience to 10 and total epochs to 50, then check gross additive spread sum. The proposed falsification criterion is a change of less than 10%.

Research priority

Continue reading papers and searching for alpha models. Keep the top three models by gross additive spread sum. Leave unresolved validation questions for later, with a friend handling validation if possible.